

BID-ASK SPREAD FORMULA AND LIQUIDITY COST RISK AND REWARD OF A MARKET MAKER

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ABSTRACT

One of the fundamental properties of most markets is the existence of more than one asset price or distribution of prices for a given asset. In many markets this distribution is clearly bi-modal, and can be related to the so-called “bid” and “ask” positions. The underlying detailed “microscopic” kinetics of bids, offers, and their matches can be complex and includes information flow along with capital and inventory balances. The reduction of a micro-model to the observed bid-ask spread determines the parameters of interest. We study a dynamic modification of the Garman model without inventory shortages, and obtain a relationship between the bid-ask spread, Value-at-Risk of the market maker, required returns, and the rate of arrivals of orders (measure of liquidity). Our formula for the bid-ask spread can be parametrized by historical data and used as a key ingredient in the information systems and/or automatic trading systems. The formula is derived in the risk-premium pricing framework suggested earlier by one of us and Guo. The bid-ask spread and liquidity costs are computed theoretically and compared with estimates for the most traded equity stocks. The remainder of the paper explores the benefits of dynamic bid-ask trading strategies.

Keywords: microeconomics, auctions, Garman model, bid-ask spread, risk premium
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Of the maxims of orthodox finance, none, surely, is more anti-social than the fetish of liquidity, the doctrine that it is a positive virtue on the part of investment institutions to concentrate their resources on the holding of "liquid" securities.

J. M. Keynes ^[1]

1. Introduction

Liquidity is closely related to price. The creation and operation of exchanges, financial institutions, contract tailoring, e-trading, literature in private, and public domain are targeted at maintaining liquidity. The idea of market efficiency and the possibility of dynamic risk control are usually associated with the market ability to quickly absorb even large size orders or neutralize deviations from equilibrium price. Such capacity comes at a cost. It is the purpose of this paper to derive how much a market maker can gain or lose while maintaining the market at a certain exchange rate. The cost of this risk is transferred to the exchange participants and may be implicitly formalized in the market rules.

The study of market microstructure currently proceeds along the statistical and information flow directions ^[2]. Before the seventies, the price formation was envisioned as an equilibrium process with an infinite number of position modifications in finite time, if needed. In this process buyers and sellers submit their order propositions, and an imaginary auctioneer communicates an appropriate average price, e.g. a price at which the auctioneer net cash position is zero. When the price is available the buyers and sellers modify their propositions and communicate them to the auctioneer. The cycle repeats as many times as needed before a single asset is transacted. By assumption, this is a converging process, and its details are irrelevant.

Beginning with the work by Demsetz ^[3] (see also ^[4]), the existence of different bid and ask prices or transaction costs has been intensively addressed. The cost was implicitly related to the cost of time needed for negotiation. Two equilibria were proposed: one for buyers, and one for sellers, even in the absence of a market maker. In an exchange dominated by market makers, where one can get the bid and ask quotes separately, and orders arrive intermittently, the bid-ask spread may diminish if the maker has sufficient inventory and capital. The quotes in liquid exchange get revised quickly and are relatively less dependent on the transaction size. From the viewpoint of outsiders the stabilizing service offered by market makers has a clear benefit measured by the smaller bid-ask spread and shorter transaction times. Despite the existence of temporal diversification the capital flow to the market maker is, on average, positive and grows with liquidity. This cost and its relationship to the bid-ask spread are studied here.

Current literature along the statistical alley is mostly devoted to the existence of stationary distribution in the time series of transactions. We refer to the paper by Domowitz and El-Gamal ^[5] for recent development in this area. The distribution is contingent on the behavior of market makers and the arrival/migration statistics of bids and asks. Surely, if, say, offers suddenly cease to arrive for a long enough time, the spread distribution may exhibit a non-stationary broadening. Similarly, if market makers

receive additional capital or larger Value-at-Risk after a successive trading session, the bid-ask spread may diminish.

In this (also statistical) paper we begin with a microscopic model of exchange, and then “coarse-grain” it to relate the spreads with risk and return over a specified time horizon, and the rate of arrival of orders. This rate is, once again, our adopted measure of liquidity. The conventional name of this rate is “volume” which usually refers to the number of traded contracts per day. If available, this measure can be easily related to other measures such as the fraction of the shares on the market which exchange hands per day or typical time needed for a given transaction. The focus of our model, described in Section 2, is on the capital of the market maker, and not on the inventory. Proliferation of e-trading seems to lead to a situation where “physical delivery” of the bought and sold assets is not really a part of any transaction, in the same spirit as it happened with the gold standard earlier in the XXth century. One may argue that companies providing e-trading services have to take offsetting positions on the market, and the real market maker has to deal with the inventory. In commodity markets it is namely the physical delivery that may shape demand and supply close to contract expiration. Nevertheless, at the current rate of proliferation of cash-settled e-trading, one may expect that soon e-clients will be only buying and selling certain claims, merely contingent on the “spot” prices of non-virtual shares. Liabilities of the e-market maker may be partially covered by securitization. In this case economics of the “assets”, i.e. intermittent cash flows arising from e-trading with the clients, have to be studied separately. Interaction of momentum and contrarian traders, intra-day and buy-and-hold traders may boost this structuring process.

Microscopic description of asset exchange is introduced in Section 2 where the ordering process and matching are explicitly modeled. The key dynamic variables include bid and ask positions at different prices and capital of the market maker. In its simplest version with static bid and ask prices, Sub-Section 2.2, our model is then simpler than the Garman model ^[6]. Similar to Garman, who studied the probability for the market maker to run out of capital or inventory, we use the profit and loss probability distribution for the maker’s capital to relate the returns with the Value-at-Risk. This allows one to derive the formula for the bid-ask spread based on observable parameters, without recourse to a utility function. This formula is discussed in detail in Section 3 where Table 1 shows how the cumulative cost of liquidity grows with traded volume (not the cost per contract).

The assumed monopoly of the market maker(s) allows him/her(them) to make abnormal returns, and may, in turn, if a capital support can be found elsewhere, trigger additional competition mechanisms, not modeled in this paper. Our problem is to find the strategy corresponding to a specified return on capital, reflecting the VaR restrictions. Such strategy will serve as the marginally “safe” market making, when the risk is mitigated to the extent it is based on VaR, and statistical expectations. If somebody consistently quotes smaller spreads he/she may run out of capital much quicker: this may be thought of as “statistical defense against irrational competition”. Other participants with capital support may challenge the static strategy by inventing dynamic ones, e.g. organize a run on the market maker(s). Such things, once again, are outside of the framework of this paper, with the exception of Section 4. Our technique can be adapted to handle multiple competitive makers with different capital support, and the resulting “darwinian” evolution. Once again, we are far from offering market maker(s) monopoly as a model of exchange. We think that modeling a statistically marginal exchange provides an

important benchmark for any exchange. There are always additional strategies on the market, which may be beyond the scope of the model, such as group formation or runs in our case. Our major philosophy here is to argue that while *very low liquidity is unacceptable, very high liquidity is too expensive*, by relating the exchange volume to the economics of the underlying assets.

2. Microscopic Description of Asset Exchange

Consider a liquid traded asset characterized by *stochastic* arrival processes, $S_a(F, t)$ and $S_b(F, t)$ of sell and buy orders, correspondingly. The rates $S_{a,b}$ measure the size of the incoming orders per unit time (number of assets per second). Rates of arrival depend on asset price, F ($0 < F < \infty$), and time, t ¹. We assume that the market maker is always prepared to quote, and perform the transaction (at the quoted price). Once confirmed, the bid and ask orders are then placed on the maker's balance sheet. The microscopic kinetics of this book is of interest for us. Removal of a given order then becomes an independent transaction, and its remaining correlation with the initial arrival (in volume if not in price) is dismissed. The kinetic equations for the occupation numbers of assets and liabilities on the maker's book of business, $n_{a,b}(F, t)$, read

$$\frac{\partial n_a(F, t)}{\partial t} = S_a - M_a, \quad \frac{\partial n_b(F, t)}{\partial t} = S_b - M_b, \quad (2.1)$$

where $M_{a,b}$ -are matching rates representing the process of taking orders off the book of the market maker. The maker has controls – supports of $S_{a,b}$, at a given moment in time, i.e. the maker is prepared to book additional client offers or bids only at a specified price. Given the instantaneous occupation numbers $n_{a,b}$, the matching rate is entirely under the maker's control, as the liability to find the counter-party is completely transferred to the market maker in this model.

The arrival rates consist of stochastic time series of order volumes at the prices specified by the market maker,

$$S_{a,b}(F, t) = s_{a,b}(F, t) d(F - F_{a,b}(t)) \quad (2.2)$$

where $F_{a,b}$ are the ask and bid prices, and the “amplitudes” $s_{a,b}$ (smooth functions of F) are characterized by their (historical) distributions, $p(s)$ ². Here $d(x)$ is the delta-function.

¹ It may also depend on the entire historical information. This latter dependence is then projected onto (F, t) plane. The systematic part is modeled as deterministic, and the residual is assumed to be stochastic (see below). We are concerned with the market microstructure, not with forecasting.

² The transition matrix of observing a bid or ask at a certain price provided the previous ask or bid occurred at another specified price^[2,5] is not used. We assume that the details of the exchange microstructure are not really known to buyers and sellers. This is

Matching takes place in the following manner. Suppose that at a given moment the market maker has only bids on his book. Usually, when an offer is booked, it is matched with the highest bid, and vice versa. If the volume of the highest bid is insufficient, the next (lower) bid is used, down to a certain boundary (hopefully, still above the offer price). Suppose matching takes place between booked buy and sell orders at prices F', F , respectively. If matching takes place at the same price, $F = F'$, the market maker makes no money ³. If $F > F'$ (or $F < F'$) the market maker has a profit (or loss). The usual positive bid-ask spread corresponds to positive support, $F > F'$. Transactions at negative $F < F'$ may occur when the market maker is trading at a loss to avoid larger losses in the future developing, for example, from an unexpected bias in bid and ask arrivals. Despite the presence of all three controls (supports of $S_{a,b}$ and matching function M) the dynamics of the system is stochastic contingent on exogenous variability of arrivals, $S_{a,b}$. As time goes on, the price time series may lose its autocorrelation and become markovian.

2.1 The Matching Term

We will model the matching process using the following approximation. Matching is performed at a certain very high rate, at first, between bids and offers with the largest difference in their prices. Then the next largest pairs are removed, et cetera, as long as the prices lie outside of the specified price window. The matching term reads

$$\begin{aligned} M_a(F, t) &= \frac{1}{t} \int_0^{F-x_0(t)} dF' d(F'-F_b^+(t)) d(F-F_a^-(t)), \\ M_b(F, t) &= \frac{1}{t} \int_{F+x_0(t)}^{\infty} dF' d(F-F_b^+(t)) d(F'-F_a^-(t)), \end{aligned} \quad (2.3)$$

The price $F_a^-(t)$ is the lowest price where $n_a(F, t)$ is positive, and $F_b^+(t)$ is the highest price where $n_b(F, t)$ is also positive. x_0 is the minimal price window used by the market maker, and time t is assumed to be small when compared with the typical scales of evolution of occupation numbers. The sole purpose of this approximation is to present the matching term in analytical form.

Using the specified arrival and matching terms (2.2), (2.3-4) the kinetic equations of the model read,

certainly not the case in exchanges without market makers, and may be partially violated even in the presence of more than one maker. However, we find it useful to model pure cases before addressing a hybrid situation.

³ Expense provisions are disregarded here and should be charged on top of the derived spread.

$$\begin{aligned}
 \frac{\partial n_a(F, t)}{\partial t} &= s_a(F, t) d(F - F_a(t)) - \frac{1}{t} \int_0^{F - x_0(t)} dF' d(F' - F_b^+(t)) d(F - F_a^-(t)), \\
 \frac{\partial n_b(F, t)}{\partial t} &= s_b(F, t) d(F - F_b(t)) - \frac{1}{t} \int_{F + x_0(t)}^{\infty} dF' d(F - F_b^+(t)) d(F' - F_a^-(t)),
 \end{aligned} \tag{2.4}$$

System (2.4) of stochastic kinetic equations is studied below.

2.2 “Frozen” Controls

The simplest case corresponds to “frozen” controls, $F_{a,b} = \text{const}$, $x_0(t) = F_a - F_b = \text{const}$. This choice severely limits the model, as it eliminates the loss mitigation activity by the market maker and the “breathing” of the bid-ask spread. The occupation numbers become $n_{a,b} = n_{0a,0b} d(F - F_{a,b})$ and Eqs(2.4) read

$$\begin{aligned}
 \frac{\partial n_{0a}(t)}{\partial t} &= s_a(t) - \frac{q^+[n_{0a}(t)] q^+[n_{0b}(t)]}{t}, \\
 \frac{\partial n_{0b}(t)}{\partial t} &= s_b(t) - \frac{q^+[n_{0a}(t)] q^+[n_{0b}(t)]}{t}.
 \end{aligned} \tag{2.5}$$

Here the right Heaviside function is defined as

$$q^+(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

The solution of (2.5), in the limit $t \rightarrow 0$, is given by

$$\begin{aligned}
 n_{0a}(t) &= \left[\int_0^t dt' (s_a(t') - s_b(t')) \right]^+, \\
 n_{0b}(t) &= \left[\int_0^t dt' (s_b(t') - s_a(t')) \right]^+,
 \end{aligned} \tag{2.6}$$

provided that $n_{0a}(0) = n_{0b}(0) = 0$, and $(x)^+ \equiv \max(x, 0)$. If the processes $s_{a,b}$ have short memory, and distributions $p_{a,b}(s_{a,b})$ are stationary with well-defined cumulants, the statistical properties of (2.6) are straightforward. Consider, for example, a set of Poisson arrivals^[7] with frequency-volume distribution, $(l_{a,b}(u), n_{a,b}(u))$, parametrized by the

dummy variable u . Then the cumulative difference of arrivals in a long run is simply a normal walk with mean and standard deviation given by ⁴

$$\begin{aligned} mt &= t \int du [l_a(u)n_a(u) - l_b(u)n_b(u)], \\ s t^{1/2} &= t^{1/2} \sqrt{\int du [l_a(u)n_a(u) + l_b(u)n_b(u)]}. \end{aligned} \quad (2.7)$$

Clearly, the distributions of $n_{0a,0b}$ correspond to the positive and negative support of this gaussian distribution.

We focus below on the balance, X , of the trading account, the principal concern of the market maker. The balance is defined as the number of matched pairs, times the difference between bid and ask prices, minus the processed number of unmatched ask-orders times their prices, plus processed number of bid-orders times their prices, as booked,

$$X(t) = x_0 \int_0^t dt' M(t') - n_{0a}(t)F_{0a} + n_{0b}(t)F_{0b}. \quad (2.8)$$

Using Eqs(2.6) one finds:

$$\begin{aligned} X(t) = x_0 \min & \left[\int_0^t dt' s_a(t'), \int_0^t dt' s_b(t') \right] - \\ & - F_{0a} \left[\int_0^t dt' (s_a(t') - s_b(t')) \right]^+ + F_{0b} \left[\int_0^t dt' (s_b(t') - s_a(t')) \right]^+. \end{aligned} \quad (2.9)$$

Below, the statistical description of cumulative orders, $\int_0^t dt' s_{a,b}(t')$, assumes the form $m_{a,b}t + s_{a,b}W_{a,b}(t)$, where drift and diffusion coefficients can be related to the Poisson model, if needed, as shown above, and Wiener processes, $W_{a,b}$ are uncorrelated, for simplicity. Then one finds for the distribution of $X(T)$,

$$P(X, T) = \frac{1}{\sqrt{2pT(F_{0b}^2 s_b^2 + F_{0a}^2 s_a^2)}} \exp \left\{ -\frac{[X + (F_{0b}m_b - F_{0a}m_a)T]^2}{2T(F_{0b}^2 s_b^2 + F_{0a}^2 s_a^2)} \right\} \quad (2.10)$$

⁴ In real liquid markets the Poisson model is far from being the number one choice: typical daily volume is in the range of $10^6 - 10^7$ shares, and *many* people are trading them. The variability of this number is of the order of the number itself, which is inconsistent with one-parameter Poisson arrivals (one would need $l(u)$ and $n(u)$ with “fat tails”), hence a simple gaussian model may be more adequate for our purposes here. Real world volume distributions have fat tails.

The quantile positions, $V(q)$, on the loss side of the distribution (2.10), read

$$V(q, T) = (F_{0b}m_b - F_{0a}m_a)T - V_q \sqrt{T(F_{0b}^2 s_b^2 + F_{0a}^2 s_a^2)}, \quad (2.11)$$

where $V_q = |N^{(-1)}(q)|$ is the inverse standard normal distribution at the specified quantile, q . For example, $V_{0.99} = 2.326$. The time dependence of V_q has a weak logarithmic correction. Indeed, if the default rate is h , the cumulative probability of survival is $q = e^{-hT}$. At times when the defaults are rare, $hT \ll 1$, one finds, with logarithmic precision, by using the asymptotic form of the error-function,

$$V_q(T) = \sqrt{-2 \ln(hT)}.$$

If the bid and ask prices are chosen such that $F_{0b}m_b < F_{0a}m_a$, the quantile position first deteriorates, and then may gradually improve, indicating a profit: the benefit of temporal diversification with linear drift versus the square-root law of diffusion. The Value-at-Risk will be defined as the minimal position of the quantile over the time horizon,

$$\begin{aligned} VaR(q) &= \min_{t \leq T} V(q, t) = \begin{cases} V(q, T_0), & T > T_0 \\ V(q, T), & T \leq T_0 \end{cases}, \\ T_0 &= \frac{V_q^2 (F_{0b}^2 s_b^2 + F_{0a}^2 s_a^2)}{4 (F_{0b}m_b - F_{0a}m_a)^2} \end{aligned} \quad (2.12)$$

If $-VaR(q)$ is the amount of capital which keeps the market maker solvent, and the required rate of return on this capital is $\bar{m}$, one obtains

$$\frac{-VaR(q) + (F_{0b}m_b - F_{0a}m_a)T}{-VaR(q)} = 1 + \bar{m}T \quad (2.13)$$

Denoting $F_{0a} = \tilde{F} + \frac{1}{2}x_0$, $F_{0b} = \tilde{F} - \frac{1}{2}x_0$, where $\tilde{F}$ is the mid-price, and assuming $x_0 \ll F_{0a,0b}$, one finds

$$F_{0a} - F_{0b} = \frac{2\tilde{F}}{(m_b + m_a)} \left(-(m_a - m_b) + \frac{V_q \bar{m}}{1 + \bar{m}T} \sqrt{T(s_a^2 + s_b^2)} \right) \quad T < T_0, \quad (2.14)$$

and

$$F_{0a} - F_{0b} = \frac{2\tilde{F}}{(m_b + m_a)} \left(-(m_a - m_b) + \frac{V_q}{2} \sqrt{\frac{\bar{m}(s_a^2 + s_b^2)}{1 + \bar{m}T}} \right) \quad T \geq T_0 \quad (2.15)$$

which indicates that $T_0 = 1/3\bar{m}$, and that the case $T < T_0$ is more applicable (even with the high return of $\bar{m} = 30\%$ per annum, T_0 exceeds one year). In deriving (2.14-15) we used the pricing framework by one of us and Guo^[8].

If the market maker bears the risk of not clearing the contract till its maturity, e.g. in option trading, the risk and reward relationship (and the bid-ask spread) should reflect this additional risk. One has to add a term $(n_a - n_b)x$ to the balance (2.8), where x is the payoff (payout) of the contract at maturity, and repeat the derivation in this section.

3. Properties of The Static Bid-Ask Spread (2.14)

In deriving (2.14) it was assumed that the market maker's only occupation is to run the exchange, and generate a sufficient rate of return to attract capital. As a result, the correlation with the market or other lines of business does not enter Eqs(2.14-15). If return calculations are performed on a (market) portfolio basis, the correlation of the distribution (2.10) with the portfolio, r , becomes important^[8]. Expression (2.14) is then modified

$$F_{0a} - F_{0b} = \frac{2\tilde{F}}{(m_b + m_a)} \left(-(m_a - m_b) + \frac{V_q r \bar{m}}{1 + \bar{m}T} \sqrt{T(s_a^2 + s_b^2)} \right), \quad T < T_0. \quad (3.1)$$

From (3.1) one can estimate the benefit of portfolio-based economics.

Let us consider the bid-ask spread Eqs(2.14, 3.1) in more detail. First we observe that systematic imbalance of bid-ask rates of arrival, $m_b - m_a$, would lead to a steady profit or loss (P&L) rate, and this is compensated by the spread. Then, there is always positive second term, with the square root, which represents a variability charge. While the imbalance term is time-independent, the variability charge increases with the time horizon at small $T \ll T_0$ as $\sqrt{T}$. On a semi-quantitative level,

$$\frac{\text{Bid - Ask Spread}}{\text{Asset Price}} = \frac{(\text{Systematic Imbalance Rate}) + (\text{Non - Systematic Variability})}{\text{Average Arrival Rate}}. \quad (3.2)$$

Clearly, the average arrival rate is the measure of liquidity here, and the bid-ask spread is inversely proportional to this rate. Relation (3.2) applies to exchanges where namely the market maker bears the risks of not finding the counter-party for completing the transaction.

Let us compare these results with the results of other models. Asymptotically, at long times, the P&L distribution moves to systematic profits provided $F_{0b}m_b < F_{0a}m_a$. This

is equivalent to Garman constraint $F_{0b} l_b < F_{0a} l_a$ in our notations. Garman's second constraint, $l_a > l_b$, which guarantees a systematic component of the inventory raise of the market maker, is disregarded here, as explained above.

Similar to our version with frozen controls the bid-ask spread of the type (3.2) arises in most statistical models, such as Ho-Stoll^[9] model or O'Hara-Oldfield model^[10]. The distinct feature of our approach is that we didn't use the risk-neutral setting and utility functions. They are replaced by risk premium and return on capital considerations.

Equations in this section are derived in continuous time. However, the modification to block trading is straightforward. Suppose, in addition to the considered processes, there is a bid block of the size L_b , or offer block of the size L_a . Eq(3.1) then reads,

$$F_{0a} - F_{0b} = \frac{2\tilde{F}}{(m_b + m_a)} \left(-(m_a - m_b) - \frac{L_a - L_b}{T} + \frac{V_q r \bar{m}}{1 + \bar{m}T} \sqrt{T(s_a^2 + s_b^2)} \right) \quad (3.3)$$

and shows the response of the bid-ask spread to the block trade (provided that we still do not run into an inventory shortage). One can argue that an offer block should not necessarily increase the spread, but rather change the ask price, and, similarly, a bid block may increase the bid price instead of decreasing the spread (see next subsection). In the version of the model with frozen controls, we have only one (static) number, $x_0 = F_{0a} - F_{0b}$, to respond to dynamic environment, and (3.3) is such a response. Clearly, the bid-ask-spread response to the block trade is small if liquidity is large, as it should be: the yields for large issuers are usually smaller than for the small issuers. Note that the present model does not take into account any non-linearity of the market feedback associated with block trades.

The model can be immediately modified to take into account the situation where the market maker is partially in competition with other makers/participants and cannot fully control the price. To do so one should simply add the log-price variance (price volatility squared) under the square root in (3.3).⁵ This is the mixed case, where all sources of randomness work against the maker. In the limit opposite to what is considered here, when there is no problem with offsetting positions, but participants have little or no influence on the price at all, only the squared price volatility remains present under the square root.

3.1 The Cost of Liquidity

We are in a position to begin the discussion of the cost of liquidity. Consider two markets with the same time horizon T , where frequency of arrivals, l , increases in a times. Using (2.7) one observes that the mean rates m are then up in a times, while standard deviations s increase in $\sqrt{a}$ times. Using (3.2) we find that the systematic part of the bid-ask spread does not change, while the non-systematic charge diminishes in

⁵ All the pair-wise covariances among the bid, ask, and price processes should also be added.

$\sqrt{a}$ times. Therefore, the cumulative amount of money flowing to the market maker increases, on average, in a times for systematic part, and in $\sqrt{a}$ times for non-systematic part, as seen from (2.14). Increase of order size in a times raises costs in the same fashion.

The situation improves if the time horizon T can be diminished ⁶. Then (2.10), (2.12) suggest $T \propto 1/a$, and the non-systematic part of the bid-ask spread now diminishes in a times. On a cumulative basis (accumulating over unchanged horizon) the amount of money flowing to the market maker increases, on average, in a times for systematic part, and remains unchanged for the non-systematic part. Marginally, only in this latter case liquidity comes at no additional cost. Surely, if the cost is determined *per traded contract*, i.e. proportionally, it does not change with liquidity for the systematic part, and decreases inversely with liquidity for the non-systematic part (provided that the time horizon is scalable). The proportional measure of cost is useful when the total number of participants grows, as in public e-trading. This paragraph is summarized in Table 1.

1	Time Horizon	T	T/a
2	Systematic Part Related to Imbalance Rate	1	1
3	Non-Systematic Part Related to Variability of Orders Arrival	$1/\sqrt{a}$	$1/a$
4	Systematic Cumulative Cost of the Exchange: Total Capital Flowing on Average to the Market Maker	a	a
5	Non-Systematic Cumulative Cost of the Exchange: Total Capital Flowing on Average to the Market Maker	$\sqrt{a}$	1
6	Systematic Cumulative Cost of the Exchange: Total Capital Flowing on Average to the Market Maker Per Contract	1	1
7	Non-Systematic Cumulative Cost of the Exchange: Total Capital Flowing on Average to the Market Maker Per Contract	$1/\sqrt{a}$	$1/a$

Table 1. Changes in Bid-Ask Components and Exchange Costs Associated with the a -Times Increase in the Order Arrival Rates; In Cases 4,5 the Cumulative Costs are Computed Per Original Horizon

3.2 Deterministic Part of the Cost and Market Sensitivity

The systematic part can be minimized by setting mid price so that $m_a = m_b$ ⁷, which is beyond the scope of the model with “frozen” controls: the response of $s_{a,b}$ to the price dynamics is not considered.

⁶ Provided that this is permitted by regulations/risk management guidelines.

⁷ Consistently keeping $m_a < m_b$ will lead to inventory shortages ^[6]. It corresponds to a strategy where the market maker should be prepared to deal with the avalanche of

One the other side, it is frequently the case that there is a negligible or even negative sensitivity of the order arrivals to the micro-scale price movements. This certainly happens when the long-term or contrarian investors dominate, respectively. As a result, there is hardly a universal dynamic strategy. If the sensitivity can be defined little can be said beyond the *marginal* requirement to adjust the mid-price to keep the average arrival rates equal, $m_a = m_b$, and set up the minimal bid-ask spread to be equal to

$$F_{0a} - F_{0b} = \frac{2r\bar{m}\tilde{F}V_q\sqrt{T(s_a^2 + s_b^2)}}{(m_b + m_a)(1 + \bar{m}T)}. \quad (3.4)$$

The sensitivity requires continuous verification as the composition of investors is frequently an unknown factor. Experienced brokers have a toolbox of sensitivity tests, such as intentional price changes, jumps, trends, oscillations, price changes for preferred clients, and a rainbow of information-based methods⁸.

In is instructive to evaluate numerically the bid-ask spread and liquidity costs. It is unclear whether optimal liquidity really exists, yet raising liquidity should have a natural limit. We first briefly mention relatively illiquid markets, such as Property/ Catastrophe Services (PCS) Options traded on the Chicago Board of Trade, <http://www.cbot.com/mplex/quotes/setl/pcscls.htm>. The daily volumes are single/double digits, and may be zero. Bid-ask spreads vary, but may exceed 100%. Such markets would benefit from a monopolistic maker willing to provide capital, e.g. using the approach presented here.

Let us compute our marginal bid-ask spread (3.4). Assuming for simplicity $m = m_a = m_b$, $s = s_a = s_b$, and using values $\sqrt{s^2 T} = \frac{1}{2} mT$, $\bar{m} = 30\%$, $r = 50\%$, $q = 99\%$, $T = 1/252$ yrs one finds $(F_{0a} - F_{0b})/\tilde{F} = 7$ bps. This is the minimal spread to be charged if somebody would decide to provide capacity for the entire market of a given to-be-liquid asset. Typical market spread of mostly traded stocks is about $\frac{1}{16}$ at the market price of 100 per share, which is the same number. The result implies that there is tight room for the monopoly game. It also implies that, in such markets, the market maker might be at a disadvantage by accumulating orders without first finding the corresponding counter-party, and relaying the risk. That is why the securitization of public e-trading is far from being trivial as far as economics is concerned.

We have also studied the market compliance with Eq(3.4), and found considerable fragmentation. In detail, we selected at random a number of securities using stocks, options, indices, bonds, and commodities. Table 2. The following Table shows the observables extracted with the help of Bridge Information Systems' "Bridge Station". The Table was prepared on April 27 – May 1, 2000.

liabilities for all unmatched contracts. As discussed in the introduction there exist more than one solution here.

⁸ There are also mixed information-price tools such as "error correction".

	Bridge Symbol	Bid-Ask Spread over Market Price, bps	Volume Volatility
Stocks	MSFT	9	0.93
	ORCL	8	1.08
	KO	14	1.28
Options	AOE EH_CO	245	8.87
Bonds	LXTY%10	2	0.82
	LXTY%30	1	1.67
	TB%03	35	4.47
	TB%06	8	13.33
	ODP%08%	145	3.56
Commodities	CL.1	200	2.06
	HG.1	19	1.03

Table 2. Bid-Ask Spread Divided by Market Price versus Volume Volatility for Securities Chosen at Random to Represent Different Market Sectors.

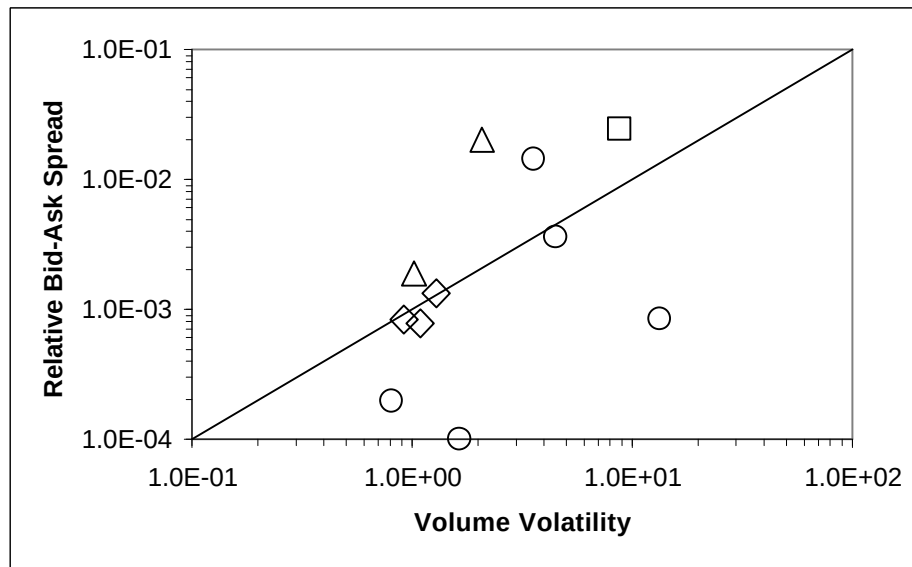


Figure 1. Data from Table 2. Stocks are Shown by Diamonds, Option by Square, Commodities by Triangles, and Bonds are shown by Circles. The Solid Line Shows the Linear Relationship of Eq(3.4).

As one can see from Table 2 and Figure 1 the data are scattered. While frequently traded stocks display rough agreement with Eq(3.4), and the option is not far from it, the bid-ask spread for two selected commodities grows roughly in proportion to variance, while bid-ask spread for bonds is “all over the chart”, and may indicate that their trading in some cases is highly “proprietary”. Addition of price volatility (see Section 3 before the

beginning of subsection 3.1) does not change the situation significantly, as the volume volatility is already very large.

We finally focus on the total amount of capital generated by a certain class of assets, $h j \tilde{F} T$, where h is the expected return rate over the time horizon T (including dividends, if any), and j is the number of shares on the market. One can compare the cost of liquidity to this amount of money and define a marginal fractional liquidity,

$$\frac{mT}{j} = \frac{hT\tilde{F}}{F_{0a} - F_{0b}} = \frac{mhT(1 + \bar{m}T)}{r\bar{m}V_q \sqrt{s^2 T}}. \quad (3.5)$$

For assets with bid-ask spread of 7 bps the right-hand side of this expression at $h = 10\%$ evaluates to 0.56. Thus, if 56% of the issued assets exchange hands every trading day, all the profit from the market appreciation of the assets will essentially be used to support the liquidity⁹. The typical market liquidity is significantly less than 56% and for the most traded stocks it is 0.5-2% on average. This doesn't seem to be excessive in terms of market value, but it is still high in terms of earnings for frequently traded stocks. See Table 3 which was prepared on March 22, 2000, and the numbers are somewhat different from those in Table 2.

Ticker	Market Cap, MM	Price, \$ as of 3/24/00	STD Volume/Average Volume	P/E Ratio	Volume, M as of 3/24/00	Annual Liquidity Cost, MM	Annual Liquidity Cost/Annual Profit	Annual Liquidity Cost * (P/E) / Annual Profit
AMTD	4	24	75%	N/A	11	0.17	43.3%	N/A
EGRP	9	31	69%	N/A	19	0.30	33.3%	N/A
CPQ	48	29	62%	97	33	0.52	10.8%	10.50
QCOM	103	145	69%	189	23	0.36	3.5%	6.65
YHOO	100	191	67%	692	6	0.09	0.9%	6.54
AOL	163	72	48%	264	17	0.27	1.6%	4.34
CSCO	536	79	35%	179	58	0.91	1.7%	3.05
ORCL	248	87	63%	157	24	0.38	1.5%	2.39
SCH	51	62	58%	89	8	0.13	2.5%	2.20
DELL	145	56	49%	82	19	0.30	2.1%	1.69
WCOM	120	43	58%	32	34	0.54	4.5%	1.43
INTC	467	139	41%	160	20	0.32	0.7%	1.08

⁹ Similarly, if only 1% of the shares exchanges hands per day and P/E ratio is 40, the same conclusion applies to the earnings. For some stocks the P/E ratio may exceed a thousand, implying, again, that the enormous liquidity is supported primarily by buyers and sellers at their own expense for their dynamic strategies if not for the pleasure of short-term gambling. See Table3.

MSFT	588	112	42%	69	48	0.76	1.3%	0.89
T	186	58	45%	27	17	0.27	1.4%	0.39
C	202	60	45%	21	12	0.19	0.9%	0.20
MO	48	20	69%	6	9	0.14	3.0%	0.18

Table 3. Cost of Liquidity for Some Frequently Traded Assets and its Relation to Annual Profit at $h = 10\%$ Return. High Liquidity Cost is not Necessarily a Drawback for Investors: Cisco Systems, Inc. Market Price Increased by a Factor of 3 over the Last Year. It is assumed that the bid-ask spread has the smallest possible non-zero value, the tick size 1/16.

Our finding is in accord with the following citation: “One possible cause of excessive trading is overconfidence. Overconfident investors will trade too frequently, that is, the gains overconfident investors realize through trade will be less than they anticipate and may not even offset trading costs. By analyzing trading records for 10,000 accounts at a large discount brokerage house, I test whether the securities these investors purchase outperform those they sell by enough to cover the costs of trading. I find the surprising result that, on average, the securities they purchase actually underperform those they sell.”¹²

4. Superiority of Dynamic Controls

Consider now the case with instant positive feedback, or (markovian) sensitivity, aka price elasticity, when the average rate of bid arrivals decreases with the price, while the rate of ask arrivals increases with the price. Over a small change of $F_{a,b}$ in liquid markets we have

$$\begin{aligned} m_a(F_a) &= m_{a0} + m_{a1}(F_a - F_0) \\ m_b(F_b) &= m_{b0} - m_{b1}(F_b - F_0). \end{aligned} \quad (4.1)$$

With linear dependencies (4.1) there is a single static solution which keeps the average rates of arrival equal.

$$\bar{F}_a = F_0 + \frac{m_{b0} - m_{a0} + m_{b1}x_0}{m_{a1} - m_{b1}}, \quad \bar{F}_b = F_0 + \frac{m_{a0} - m_{b0} - m_{a1}x_0}{m_{b1} - m_{a1}}, \quad (4.2)$$

where F_0 is an arbitrary reference price. The variability of order arrivals may also depend on the price; this is not modeled here.

Now, an interesting problem arises: is there a better dynamic strategy than to keep fixed $F_{a,b}$ with the bid-ask spread (3.4)? We do not intend to solve in full the stochastic system (2.4) with gaussian distributions of $s_{a,b}(F, t)$ having F -dependent mean values (4.1) and constant standard deviations, $s_{a,b}$, but a few steps are in order.

Consider an example of dynamic adaptive strategy where $F_{a,b}(t)$ are shifted up if the number of accumulated bids is excessive, and shifted down if the number of accumulated offers is excessive. The shift is performed in proportion to the accumulated excess,

$$\frac{dF_a(t)}{dt} = -g \left[\int_0^\infty dF n_a(F, t) - \int_0^\infty dF n_b(F, t) \right], \quad F_b(t) = F_a(t) - x_0, \quad (4.3)$$

where g is a positive constant, describing the effect of positive feedback. Given that $F_a(t) - F_b(t)$ is also a positive constant (and it is equal to the matching spread), one of the occupation numbers $n_{a,b}$ is always zero. In fact, the entire distribution is either a delta-function for ask or for bid positions. Even in this simple case we already find the dynamic strategy beneficial.

Let the parameters be symmetric, $m_{a0} = m_{b0} = m_0$, $m_{a1} = m_{b1} = m_1$, $s_1 = s_2 = s$. It follows from (4.1), (4.3), (2.4) at $t \rightarrow 0$ that the accumulated excess of bid over ask positions itself follows the so-called Ornstein-Uhlenbeck process,

$$dn = -2gm_1 n dt + s dW_1 + s dW_2, \quad (4.4)$$

which is correlated with the contribution to the market maker balance coming from order matching,

$$dx = (m + gm_1 |n|) dt + s dW_2. \quad (4.5)$$

The full balance of the trading account is

$$X = (F_{a0} - F_{b0})x + \int_0^\infty dF n(F, t) F \operatorname{sgn}(F - F_0) \quad (4.6)$$

Adequate description of the full balance, X is provided through the joint probability distribution, $P(n, x, t)$,

$$P(X, t) = \int_{-\infty}^\infty dn \int_{-\infty}^\infty dx d[X - (F_{a0} - F_{b0})x - F(n)n] P(n, x, t) = \int_{-\infty}^\infty dn P\left(n, \frac{X - F(n)n}{F_{a0} - F_{b0}}, t\right) \quad (4.7)$$

where

$$F(n) = \begin{cases} F_{a0} + gn, & n > 0, \\ F_{b0} + gn, & n \leq 0. \end{cases} \quad (4.7)$$

The joint probability distribution, $P(n, x, t)$ satisfies the following forward Fokker-Planck equation,

$$\partial_t P = d(n)d(t) - (m_0 + m_1 |n|) \partial_x P - 2m_1 n \partial_n P + \frac{s^2}{2} (\partial_n + \partial_x)^2 P. \quad (4.8)$$

Using integral transforms along x and t directions one obtains

$$lP = d(n) + (m_0 + m_1 |n|) ikP - 2m_1 n \partial_n P + \frac{s^2}{2} (\partial_n - ik)^2 P. \quad (4.9)$$

The solution of this equation is reducible to parabolic cylinder functions. It offers a possibility to write the answer in quadratures, and study its asymptotic forms. Straightforward analytic inversion of integral transforms cannot be done.

Here we only discuss the properties of the bid-ask spread at long times, $gm_1 t \gg 1$, and strong control, $gm_0^2 \gg s^2$, the most interesting case. Then the statistics of X is governed by x , and one can disregard the $x \leftrightarrow n$ coupling completely. As in the static case, Eq(4.9) reduces again to a gaussian form,

$$lP = d(n) + m_0 ikP - \frac{s^2 k^2}{2} P. \quad (4.10)$$

Therefore, all statements in the static case are again enforced with the only exception: the volatility is reduced in $\sqrt{2}$ times. The term $\sqrt{T(s_a^2 + s_b^2)} = s\sqrt{2T}$ is replaced by $s\sqrt{T}$ in (2.14), (2.15), (3.1), (3.3), (3.4). We were able to compensate the “difference mode” between bids and ask arrivals by using dynamic control (4.3). Note that not the result is dependent on the control parameters m_1, g but the region where difference mode is compensated. This implies that the obtained result is already optimal.

Conclusions

We have considered a dynamic model of economics of a market maker running a monopolized liquid exchange under Value-at-Risk and return on capital constraints. It was shown that the marginal dynamic strategy should better maintain the mid-price at the level of balancing the bid and ask orders (if possible). The residual bid-ask spread is associated with uncertainty and can be estimated (priced) on a stand-alone and portfolio basis. In both cases the cumulative cost of liquidity borne by the market participant is proportional to the square root of liquidity itself, the latter being measured by the daily volume of exchanged shares. Simple dynamic strategies offer an advantage over the static approach, provided that the market sensitivity is known. To our understanding, the description of dynamic strategies in bid-ask spread accumulation bears noticeable similarity to dynamic investment strategies considered earlier by one of us^[11]. The transparent nature of the results listed above allows one to design and “educate” a computer-mediated automatic trader based on principles of risk and return. Such an e-trader will work in moderately liquid markets where the bid-ask spread is of the order of

few percent of the asset price or larger. It will sample the market sensitivity, adjust the bid and ask quotes, and may become a competitor to human traders.

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